

Group Number:
Student Names:
Marks: 4 points

COMP 251: Data Structures & Algorithms
Group Assignment 2: Linked Lists



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******* INSTRUCTIONS *******

- Read each task carefully before starting. Make sure you understand the task requirements, expected output format, and any constraints mentioned.
- All programming tasks must be implemented in **Java**. Ensure that your code compiles and runs correctly using a standard Java development environment (recommendation: **BlueJ**).
- Use meaningful variable and method names to improve readability and understanding of your Java code.
- Provide clear, well-structured Java code and a corresponding screenshot of the output for each programming task.
- Where applicable, provide clear written explanations to justify your answers.
- All submitted work must be your own (or your group's) original work. Any form of plagiarism or unauthorized collaboration will be dealt with in accordance with university policies.

Task 1: SinglyLinkedList Class

In the lecture, we studied in detail the implementation of the Singly Linked List by reviewing code given in the code fragments 3.14 and 3.15 of the textbook, i.e., please refer to the class `SinglyLinkedList<E>`. The Java source code for the `SinglyLinkedList<E>` class can also be found on the course page <https://myclass.ufv.ca/>.

By using this class, do the following:

- Create a Linked List (in the main method) that has three integer elements and two strings. Note that you can choose any integers and strings.
- Find a `toString()` method in the `SinglyLinkedList<E>` class that provides string representation of the contents of the list. Use this method to print the contents of the Linked List.

For instance, one of the possible outputs should look like this:

Output:

Mixed List: (5, 7, 4, Hello, World)

Task 2: [Optional] Implement a new method in SinglyLinkedList Class

Implement and add a new method which adds a new node of type 'E' after a given predecessor (provided in a parameter), i.e., the signature of this method should be like this

```
1 private void add_after_predecessor(E e, Node<E> predecessor)
```

After implementation of this method, do the following:

- Comment out the code that is provided in addLast (E e) method in the SinglyLinkedList<E> class. Now use your implemented add-after-predecessor() method to modify the addLast (E e) method.
- Run the following main method

```
2 public class Task2
3 {
4     public static void main(String[] args) {
5         // Create an instance of the SinglyLinkedList class
6         SinglyLinkedList<Object> mixedList = new SinglyLinkedList<>();
7         mixedList.addFirst("Hello");
8         mixedList.addFirst(4);
9         mixedList.addFirst(7);
10        mixedList.addFirst(5);
11        mixedList.addLast("World");
12        System.out.println("Mixed List: " + mixedList);
13    }
14
15 }
```

The output should remain exactly same as in the Task 1, i.e.,

Output:

Mixed List: (5, 7, 4, Hello, World)